

NSLS-II. Photo: Brookhaven National Laboratory

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# Turning a Chess Engine into Science AI

In one week, 20 students build a chess AI in plain Python, then aim it at a science problem modeled on Brookhaven's labs.

| 30 | 20 | ~760 | 0 |
|---|---|---|---|
| class hours over 5 days, four 90-minute periods a day | students in 10 student pairs, coding and playing chess together | lines of Python each student pair types in and runs | student online accounts; everything runs offline after the Day 1 setup |

## Chess is where AI learned to think ahead. Science is where it is going next.

Every game-playing AI runs the same loop: describe the situation, list the possible actions, look ahead, score the results, and check the answer. An AI agent at a beamline runs that loop to choose its next measurement.

Students who know some coding and know how to play chess, including notation, build **Remus**, a complete chess engine with no hidden libraries. Each day they type in the next version: Remus1 plays legal chess, Remus2 searches ahead, and Remus3 and Remus4 add the enhancements of real engines, each tested against the version before in Lucas Chess. Students **build a model, test it against a reference, and measure the results**.

### SAME LOOP, TWO WORLDS

| | Represent | Generate | Look ahead | Evaluate | Verify |
|---|---|---|---|---|---|
| IN REMUS | An 8×8 board and the full game state | Every legal move, including en passant | Minimax and alpha-beta search | Score material and position | Perft counts match a trusted reference |
| IN SCIENCE AI | A sample, a model, or an instrument state | Candidate measurements or hypotheses | Predict outcomes before spending beam time | Score how well each hypothesis explains the data | Compare predictions with measured data |

## What students will be able to do AND HOW WE WILL KNOW

| OUTCOME | EVIDENCE A REVIEWER CAN CHECK | WHEN |
|---|---|---|
| **Model a system as data** | Remus1 holds every chess rule and plays only legal moves in Lucas Chess | Day 1 |
| **Read and reproduce real code** | Each student pair types four engines from the instructor's code and runs each file | Days 1-4 |
| **Reason about search** | Remus3 searches deeper than Remus2 in the same five seconds, shown in its info lines | Days 2-3 |
| **Measure a change** | Each version plays the one before, and Remus4 plays the same engine in Rust; results compared day by day | Days 2-4 |
| **Find a hidden peak** | Each student pair's AI finds a hidden hot spot in fewer readings than random guessing | Day 5 |
| **Name the crystals in a pattern** | On 200 known X-ray patterns, its tree search makes fewer wrong calls than a greedy guess | Day 5 |
| **Explain a result in plain words** | Each student pair presents its science results and two powders in three minutes | Day 5 |

## The decision

1. **Approve a pilot** for 20 students, designed to grow to 30: one classroom for 5 days, 8:30 to 4:30, 10 Windows 11 laptops, and 20 USB drives.
2. **Arrange a scientist visit** on Day 5, and share one measured two-phase powder pattern from the PDF or XPD beamline for the phase hunt.
3. **Support an evaluation** with Brookhaven education staff: a concept check and an interest survey before and after the week.
4. **Judge it by evidence:** student code, games, science results, and explanations, not chess strength. No student accounts; code goes home on USB.